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Hachimijiogan (*Ba-Wei-Di-Huang-Wan*), a herbal medicine, improves unbalance of calcium metabolism in aged rats

Ryuji Ikeda, Kazushige Mizoguchi*

Section of Oriental Medicine, Department of Geriatric Medicine, National Institute for Longevity Sciences, National Center for Geriatrics and Gerontology (NCGG), 36-3 Gengo, Morioka-cho, Obu, Aichi 474-8522, Japan

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ABSTRACT

Aim of the study: Aged animals as well as elderly humans commonly exhibit calcium (Ca) shortage because of increased Ca excretion into urine and decreased intestinal Ca absorption, which induce elevation of serum PTH levels to maintain serum Ca levels between a normal physiological range. The most important organ that regulates this Ca homeostasis is the kidney. *Hachimijiogan* (HJG), a traditional herbal medicine in Japan and China, has been used for treating clinical diseases associated with kidney dysfunctions in elderly humans. However, the mechanisms of its pharmacological actions remain to be understood poorly. The present study was designed to examine whether HJG improves age-related unbalance of Ca metabolism at the systemic level using aged rats.

Materials and methods: HJG was administered to 21-month-old aged rats for 3 months, and several parameters associated with Ca metabolism in serum and urine were measured.

Results: Although HJG as well as aging itself did not affect serum Ca levels compared to young (11-week-old) rats, HJG improved increase in urinary Ca excretion and elevation of serum parathyroid hormone (PTH) levels in aged rats. However, HJG did not improve marked reduction of intestinal Ca absorption in aged rats.

Conclusion: HJG showed regulating action for age-related unbalance of Ca metabolism at the systemic level. This finding would provide useful information for treating age-related several disorders associated with Ca unbalance.

1. Introduction

Aging is considered as a complex multifactorial process that results in heterogeneous patterns of progressive morbidity and disability (Rowe and Kahn, 1987; Seeman and Robbins, 1994). This process is influenced by multiple homeostatic mechanisms that include hormonal and neural systems. In addition, one aspect of aging research that has come to the forefront is thought to be age-related abnormal metabolism of minerals, e.g., calcium (Ca) (Raza et al., 2007). Ca is a ubiquitous intracellular second messenger that plays a key role in variety of cellular function including growth, intracellular signal transduction, survival, and death (Arundine and Tymianski, 2003). Clinically, reduced Ca levels in blood in elderly population results in secondary hyperparathyroidism, which in turn increases bone turnover, accelerates bone loss, and increases the risk of osteoporosis (Lips and Tymianski, 2001). Thus, aging is not only a crucial risk factor affecting overall dysregulation of physiological function, but also reduces Ca metabolism that increases the risk of age-related diseases (Lanske and Razzaque, 2007).

For many years, *Hachimijiogan* (HJG, in Japanese; *Ba-Wei-Di-Huang-Wan* in Chinese), one of traditional herbal medicines (called *Kampo* drugs in Japan, remedies composed of specified mixture of crude drugs derived from plant, animal, and mineral materials), has been used for treating "kidney deficiency" that is a concept in traditional Chinese medicine and that means reduced energy to live, dysfunctions of brain, eyesight, and hearing, and decreased bone metabolism, in addition to dysfunction of the kidney itself. According to these theories, "kidney deficiency" is thought to be involved in the aging pathophysiology. Indeed, the effectiveness of HJG has been indicated in several clinical studies. For example, HJG is effective in several kidney diseases such as renal nephritis, diabetic nephropathy, nephritic syndrome, and glomerulonephritis without toxic and/or side effects (Nakagawa et al., 2005; Yamabe et al., 2005, 2007; Yamabe and Yokozawa, 2006; Hirotani et al., 2007). Moreover, HJG has protective effects on decreased bone density such as osteopenia (Ogirima et al., 2005; Kanehara et al., 2006) and osteoporosis (Chen et al., 2005). Although scientific evidence of the clinical effects of this drug is scarce, Hirawa et al. (1996) has demonstrated that HJG improves kidney dysfunction in salt-

* Corresponding author. Tel.: +81 562 46 2311x5553; fax: +81 562 48 2373.
E-mail address: k-mizo@nils.go.jp (K. Mizoguchi).

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Table 1
Crude drug composition of *Hachimijiogan*.

Plant name	Composition (g)	Major component
Rehmanniae radix	6.0	Catalpol
Corni fructus	3.0	Loganin
Dioscorea rhizoma	3.0	Diosgenin
Alismatis rhizoma	3.0	Alisol
Poria	3.0	Eburicoic acid
Moutan cortex	2.5	Paeonol
Cinnamomi cortex	1.0	Cinnamaldehyde
Processi aconiti tuber	0.5	Aconitine

induced hypertensive rats. Alternatively, HJG has been shown to improve scopolamine-induced memory impairment (Hirokawa et al., 1996) and to have actions to maintain lens transparency by modulating unbalance of sodium, potassium, and Ca ions in mice bearing cataract (Kamei et al., 1987).

Considering that the most important organ that regulates Ca homeostasis is the kidney (Honda et al., 2007), we hypothesized that abnormal Ca metabolism occurs at the systemic level during the normal aging process, and HJG prevents this abnormality. To test this hypothesis, we designed the following experiments and investigated the effects of HJG on Ca metabolism in aged rats *in vivo*. First, we examined the effects of HJG on Ca and creatinine (Cr) concentrations in serum and urine, urine volume, and total amount of excreted Ca into urine in aged rats. Next, we tried to clarify the mechanisms underlying the pharmacological actions of HJG, focusing an important hormone that regulates Ca metabolism, parathyroid hormone (PTH), and the efficiency of intestinal Ca absorption.

2. Materials and methods

2.1. Drug

HJG was supplied in the form of a water-extracted dried powder that was manufactured from a mixture of the crude drugs listed in Table 1 (Tsumura Co., Tokyo, Japan). The concentration of several effective chemicals is defined for each crude drug as an internal standard in the company's guide to Good Manufacturing Practices.

2.2. Animals and drug treatment

All animal experiments were performed in accordance with our institutional guidelines after obtaining permission from the Laboratory Animal Committee at the National Center for Geriatrics and Gerontology. The experiments in the present study were designed to minimize the number of animals used and to minimize their suffering.

Naive male F344/N rats were used in the present study. Eleven-week-old rats weighing 190–210 g were purchased from Japan SLC (Shizuoka, Japan) and used as the young rats. Twenty-one-month-old rats weighing 380–440 g were obtained from the Aging Farm, which produces aged rats under specific pathogen free condition and was established at our institute in 2000 (Tanaka et al., 2000), and used as the aged rats. They were housed two per cage in a temperature-controlled (23 ± 1 °C) and light-controlled (12 h light/dark schedule; lights on at 8:00 a.m.) environment and were fed laboratory food and water *ad libitum*.

Young rats were fed standard pellet chow for rodent containing 1.09% Ca, 0.93% phosphate, 25.5% crude protein, and 2.5 IU vitamin D_3 (CE-2; Clea Japan, Tokyo, Japan) and used as the young control group. Aged rats were divided into two groups. One group was fed standard pellet chow for 3 months as the aged control group. The other was fed standard pellet chow containing 3% HJG for 3 months as the HJG-treated aged group.

2.3. Serum and urinary parameters

Blood were drawn from the tail vein under pentobarbital anesthesia (50 mg/kg, for young rats; 30 mg/kg for aged rats, i.p.). Urinary samples were collected from the rat housed individually in metabolic cages (Natsume Seisakusho Co, Tokyo, Japan). Ca and Cr concentrations in serum and urine were measured by commercially available kits (Wako Pure Chemical Industries, Osaka, Japan). Total amount of Ca excreted into urine was calculated using the values of urinary volume and urinary Ca concentration. Cr clearance was calculated by standard formula. PTH was measured using an ELISA kit (Immunotopics, San Clemente, CA, USA).

2.4. ^{45}Ca administration experiment

Rats were fasted overnight before the experiments. ^{45}Ca (21.6 mCi/ml; PerkinElmer, Kanagawa, Japan) diluted with distilled water was orally administered to each rat (90 mCi/kg). After the administration, 10 μl of blood samples were obtained from the tail vein during 540 min at various intervals indicated in Fig. 4 and collected in scintillation vials. To remove the quenching effect of blood materials, 40 μl of 30% hydrogen peroxide solution was added to each vial, and then 1 ml of Opti-Flore liquid scintillation cocktail (PerkinElmer) was added. Radioactivity of ^{45}Ca was measured by a liquid scintillation counter (LSC-5100; Aloka, Tokyo, Japan) with an efficiency of 99%.

2.5. Statistical analysis

All data were initially analyzed using one-way analysis of variance (ANOVA). The comparisons of the increasing response of ^{45}Ca radioactivity appeared in blood in each experimental group were made using the paired *t*-test. The comparisons of remaining data

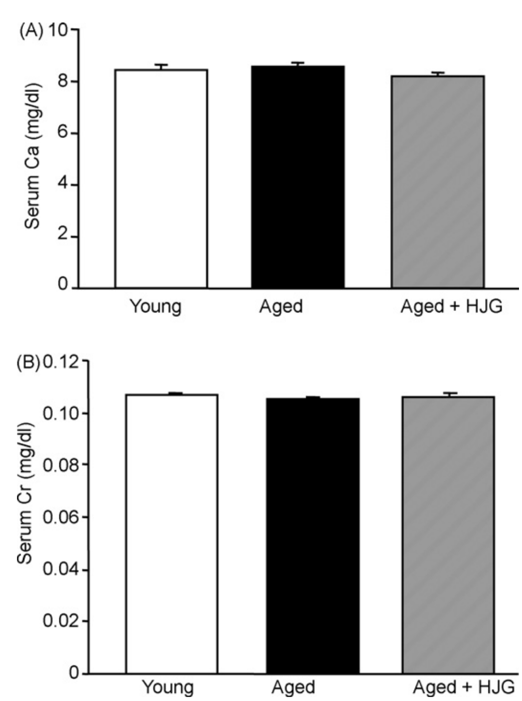

Fig. 1. Serum Ca (A) and Cr (B) concentrations in young control, aged control, and HJG-treated aged rats. Each column is expressed as the mean ± S.E.M. (young control, $n = 8$; aged control, $n = 6$; HJG-treated aged, $n = 9$). There were no significant differences in Ca and Cr concentrations between three groups.

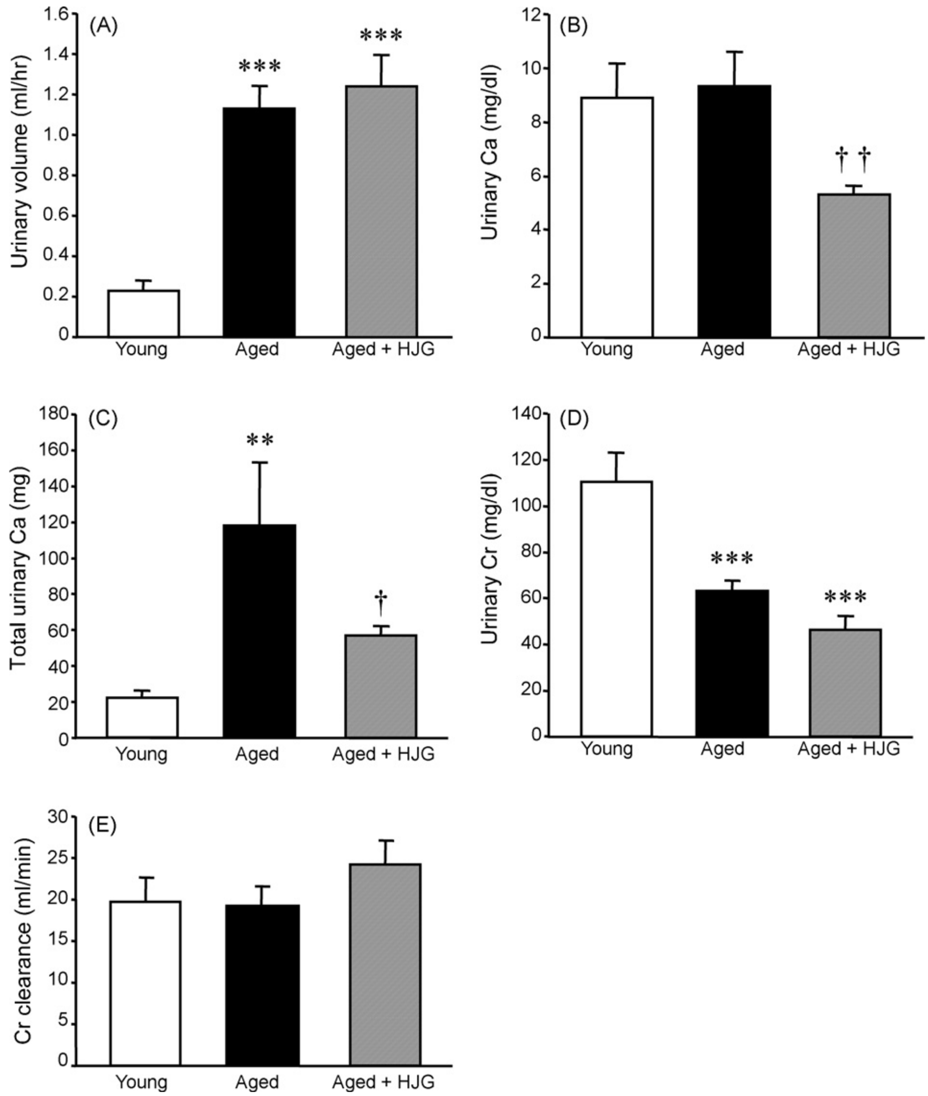

Fig. 2. Urinary volume (A), urinary Ca concentration (B), total amount of excreted Ca into urine (C), urinary Cr concentration (D), and Cr clearance (E) in young control, aged control, and HJG-treated aged rats. Each column is expressed as the mean ± S.E.M. (young control, $n = 7$; aged control, $n = 5$; HJG-treated aged, $n = 5$). $^{**}p < 0.01$, $^{***}p < 0.001$ vs. young control; $^{\dagger}p < 0.05$, $^{\dagger\dagger}p < 0.01$ vs. aged control.

were made using the Fisher's protected least significant difference test.

3. Results

3.1. Ca and Cr in serum

Ca and Cr concentrations in serum of young control, aged control, and HJG-treated aged rats are shown in Fig. 1. There were no significant changes in Ca and Cr concentrations between young and aged rats, and HJG treatment did not affect these concentrations. It is noted that there also was no significant difference in the ratio of Ca/Cr (data not shown). It is noted that there was significant difference in body weight between young and aged control rats [$F(2, 24) = 82.230$, $p < 0.001$], but HJG did not affect body weight of aged rats (Table 2).

3.2. Volume, Ca, and Cr in urine

Urinary volume, Ca and Cr concentrations in urine, total amount of Ca excreted into urine, and Cr clearance of young control, aged control, and HJG-treated aged rats are shown in Fig. 2. Urinary volume was significantly increased in aged control and HJG-treated aged rats compared to young rats [$F(2, 25) = 7.395$, $p < 0.001$, respectively] (Fig. 2A). There was no significant difference in it between aged and HJG-treated aged rats. Although Ca concentration in urine was not changed between young and aged control rats, the concentration was significantly decreased in HJG-treated aged rats compared to aged control rats [$F(2, 27) = 3.515$, $p < 0.01$] (Fig. 2B). Total amount in urinary Ca was significantly increased in aged control rats compared to young control rats [$F(2, 24) = 6.403$, $p < 0.01$], and this increase was significantly suppressed in HJG-treated aged rats [$F(2, 24) = 6.403$, $p < 0.05$] (Fig. 2C). Although Cr concentration

Table 2
Body weight of young control, aged control, and HJG-treated aged rats.

	Young	Aged	Aged-HJG
Body weight (g)	196 ± 2.6	407.5 ± 4.8***	417 ± 4.2***

Each value is expressed as the mean ± S.E.M. (young control, $n = 9$; aged control, $n = 10$; HJG-treated aged, $n = 8$).
*** $p < 0.001$ vs. young control.

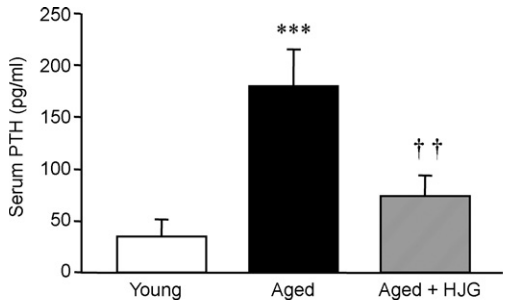

Fig. 3. Serum PTH concentrations in young control, aged control, and HJG-treated aged rats. Each column is expressed as the mean ± S.E.M. (young control, $n = 10$; aged control, $n = 10$; HJG-treated aged, $n = 10$). $^{***}p < 0.001$ vs. young control; $^{\dagger\dagger}p < 0.01$ vs. aged control.

in urine was significantly low in aged control and HJG-treated aged rats compared to young rats [$F(2, 27) = 15.693$, $p < 0.001$, respectively], there was no significant difference in it between aged and HJG-treated aged rats (Fig. 2D). Cr clearance was no significant difference between three experimental groups (Fig. 2E).

3.3. Serum PTH concentration

PTH concentrations in serum of young control, aged control, and HJG-treated aged rats are shown in Fig. 3. PTH concentration was significantly higher in aged control rats than in young control rats [$F(2, 27) = 8.749$, $p < 0.001$], and this higher concentration was significantly suppressed in HJG-treated aged rats [$F(2, 27) = 8.749$, $p < 0.01$].

3.4. ^{45}Ca radioactivity in blood after its oral administration

Radioactivity of ^{45}Ca in blood, which appeared after its oral administration, is shown in Fig. 4, and several values of Fig. 4 are picked up in Table 3. In young rats, the significant increase in ^{45}Ca radioactivity was observed from 10 min after the administration [$F(1, 14) = 5.628$, $p < 0.05$]. The radioactivity was then peaked at 80 min [$F(1, 14) = 23.149$, $p < 0.001$], thereafter, it was decreased gradually. Even at 540 min, the significant increase was observed [$F(1, 10) = 43.970$, $p < 0.05$]. In aged control or HJG-treated rats, the significant increase in radioactivity was also observed from 15 min after the administration [aged control rats, $F(1, 8) = 11.468$, $p < 0.05$; HJG-treated aged rats, $F(1, 8) = 8.538$, $p < 0.05$]. Although these radioactivities did not show the clear peak, the increased radioactivities continued up to 540 min [at 540 min; aged control rats, $F(1, 8) = 80.529$, $p < 0.001$; HJG-treated aged rats, $F(1, 8) = 26.202$, $p < 0.001$]. Comparing radioactivity between three experimental groups, radioactivity of aged control rats was dramatically low compared to young rats at many time points. As shown in Table 3, for example, radioactivity of aged rats at 80 min was significantly lower than that of young rats [$F(2, 15) = 9.337$, $p < 0.01$]. There were no significant differences in radioactivity between aged control and HJG-treated aged rats at any time points.

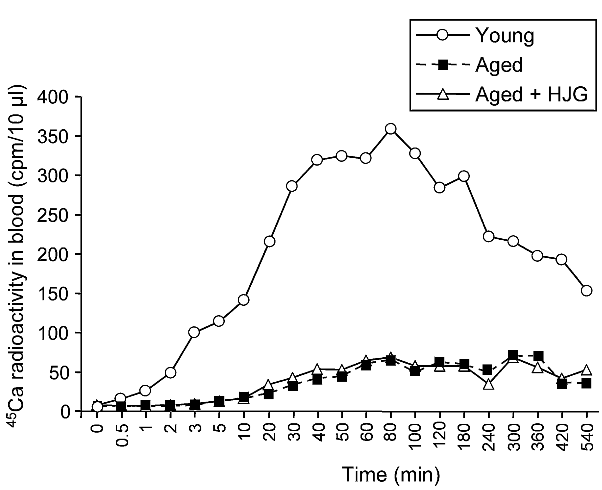

Fig. 4. Pulse-chase experiment using ^{45}Ca in young control, aged control, and HJG-treated aged rats. As a pulse, ^{45}Ca was orally administered, and ^{45}Ca radioactivity appeared in blood collected from the tail vein at several time points after the administration is plotted. Mean values of radioactivity at each time point are shown (young control, $n = 8$; aged control, $n = 5$; HJG-treated aged, $n = 5$). In detail, several values including S.E.M. are picked up in Table 3.

Table 3
Specific radioactivity of ^{45}Ca in blood after its oral administration.

Time (min)	^{45}Ca radioactivity in blood (cpm/10 µl)		
	Young	Aged	Aged + HJG
0	7 ± 0.4	7 ± 0.2	7 ± 0.3
10	$140 \pm 56^{\dagger}$	13 ± 2.3	16 ± 4.2
15	$203 \pm 72^{\dagger}$	$17 \pm 2.6^{\dagger}$	$28 \pm 6.4^{\dagger}$
30	$285 \pm 81^{\dagger\dagger}$	$33 \pm 3.3^{**,\dagger\dagger\dagger}$	$42 \pm 12.7^{**}$
80	$358 \pm 72^{\dagger\dagger}$	$65 \pm 7.5^{**,\dagger\dagger\dagger}$	$68 \pm 14.4^{**,\dagger}$
180	$298 \pm 92^{\dagger}$	$59 \pm 8.8^{**,\dagger\dagger}$	$57 \pm 11.8^{**,\dagger}$
300	$215 \pm 31^{\dagger\dagger}$	$70 \pm 10.3^{***,\dagger\dagger}$	$68 \pm 13.8^{***,\dagger\dagger}$
540	$153 \pm 32^{\dagger}$	$38 \pm 3.5^{***,\dagger\dagger\dagger}$	$52 \pm 8.9^{**,\dagger\dagger}$

Each value is expressed as the mean ± S.E.M. (young control, $n = 8$; aged control, $n = 5$; HJG-treated aged, $n = 5$).
** $p < 0.01$ vs. young control.
*** $p < 0.001$ vs. young control.
† $p < 0.05$ vs. 0 min in each group.
†† $p < 0.01$ vs. 0 min in each group.
††† $p < 0.001$ vs. 0 min in each group.

4. Discussion

In the present study, we showed that HJG has a protective action on excessive urinary Ca excretion in aged rats, presumably due to facilitation of reabsorption of filtered Ca in the kidney glomerulus, which might be involved in suppression of increased serum PTH levels.

As shown in Fig. 1, serum Ca and Cr levels did not change between all experimental groups, suggesting that Ca metabolism, at least the circulation level, is not affected during the normal aging process and HJG does not modify Ca metabolism at first sight. However, it seems that this interpretation is incorrect.

We next compared Ca excretion into urine between young control, aged control, and HJG-treated aged rats. As a result, the urinary volume was greatly increased in aged control rats (Fig. 2A), concomitant with unchanged urinary Ca concentration (Fig. 2B). This was evidenced as that total amount of excreted Ca was increased in aged rats (Fig. 2C). This increased Ca might be caused by reduced reabsorption of filtered Ca, because Ca concentration did not change in aged rats compared to young rats (Fig. 2D), despite urinary Cr concentration was lower in aged than in young, which indicates that aged urine is thin. The finding that Cr clearance did not change between young and aged rats (Fig. 2E) suggests that filtering function of the kidney glomerulus is maintained in our aged rats. This finding is consistent with a previous study showing unchanged Cr clearance in 24-month-old rats that is the same age as used in the present study; however, more aged, i.e., 28-month-old rats showed decreased clearance (Halloran et al., 2002) that is also consistent with human observations (Pattanaungkul et al., 2000). In drug treatment study, although HJG did not affect urinary volume and Cr concentration, it decreased urinary Ca concentration, which led to the result showing that HJG decreased total amount of excreted Ca into urine (Fig. 2C). Although the mechanisms underlying the improving effect of HJG are remained to be studied, the findings that HJG did not affect the urinary volume, urinary Cr concentration, and Cr clearance as well as serum Ca and Cr concentrations, but decreased urinary Ca concentration, may suggest that HJG facilitated renal tubular reabsorption of filtered Ca.

One of important hormones that regulate Ca levels in urine as well as serum is PTH. When serum Ca levels decrease, PTH secretion increases, which leads to facilitation of Ca reabsorption on the distal renal tubule and of Ca mobilization from the bone to maintain serum Ca levels within a normal physiological range. In the present study, we observed that serum PTH levels in aged rats were elevated, which is in line with human studies that serum PTH levels increase with advancing age (Wiske et al., 1979; Marcus et al., 1984; Orwoll and Meier, 1986; Forero et al., 1987; Young et al., 1987; Eastell et al., 1991). Considering the increased total urinary Ca excretion concomitant with elevated serum PTH levels, reabsorption function might become desensitization in response to elevated PTH levels in aged rats. Indeed, similar observations have been reported that the capacity of PTH to increase serum levels of 1,25-dihydroxyvitamin D [1,25$(OH)_2$D] declines with age in both rats and humans (Ambrecht et al., 2007). In addition, elevated PTH levels in aged rats might be caused by increased urinary Ca excretion to maintain constantly serum Ca levels through facilitation of Ca mobilization from the bone. Interestingly, age-related increase in PTH levels was suppressed by HJG treatment (Fig. 3), which is thought to be caused by improvement of the age-related increase in total urinary Ca excretion (Fig. 2C). Recently, Björkman et al. (2008) have shown that elevated serum PTH levels predict impaired survival prognosis in general aged population. This may be related with hypertension and/or left ventricular hypertrophy, because elevated PTH levels are associated with these diseases (McCarron et al., 1980, 1982; Nasri et al., 2004; Randon et al., 2005). Therefore, HJG may interrupt the development of these diseases. Indeed, HJG has been reported to improve hypertension and kidney glomerular injury in hypertensive rats (Hirawa et al., 1996).

An important cause of the secondary hyperparathyroidism in elderly individuals is an age-related impairment in intestinal Ca absorption (McKane et al., 1996; Riggs et al., 1998), which provokes Ca deficiency. Aging is associated with decreased intestinal Ca absorption (Wood et al., 1998). Therefore, the tracer experiment by using ^{45}Ca was performed to examine the efficiency of intestinal Ca absorption *in vivo*. All groups of rats received an oral administration of ^{45}Ca as a tracer, and measured radioactivity of administered ^{45}Ca appeared in blood that drawn from the tail vein at several post-administration time points. As indicated in Fig. 4 and Table 3, aged rats exhibited dramatically decreased ^{45}Ca radioactivity compared to young rats, indicating that intestinal Ca absorption is greatly reduced in aged rats. However, HJG had no effect on the reduced intestinal Ca absorption. This finding suggests that the suppressing effect of HJG on age-related PTH elevation is not caused by enhancement of intestinal Ca absorption.

Taken together, the present findings suggest that systemic Ca shortage mediated by increased Ca excretion in the kidney and reduced Ca absorption in the intestine occurs during the normal aging process and that HJG improves the increased excretion rather than the reduced absorption, which may cause suppression of age-related elevation of serum PTH levels. Thus, HJG did not increase an input (i.e., intestinal absorption) of Ca, but did decrease an output (i.e., urinary excretion) of it.

At present, the underlying mechanisms of the improving effect of HJG on the age-related increase in Ca excretion and responsible crude drug(s) and active ingredient(s) for the effect are unknown. HJG is composed of eight crude drugs, as listed in Table 1. Among them, *Poria* contains not only eburicoic acid, but also ergosterol that is the progenitor of 1,25$(OH)_2D_2$. Recently, it has been shown that supplementation of 1,25$(OH)_2D_3$ that has the same physiological actions as 1,25$(OH)_2D_2$ causes an increase in renal vitamin D receptors (VDRs) in mice (Healy et al., 2003), which is required for reabsorption of filtered Ca in the distal renal tubule. Similarly, Costa and Feldman (1986) have demonstrated that injection of 1,25$(OH)_2D_3$ has little effect on duodenal VDRs, while it up-regulates renal VDRs in rats. The mechanisms underlying the pharmacological actions of HJG remain to be studied.

5. Conclusion

From AD 2001 to 2026, the number of 65 years and older is predicted to increase approximately from 550 to 973 million in the world (Kinsella, 2001). The elderly population has a risk of some clinical complications, called geriatric diseases. The traditional Japanese (*Kampo*) and Chinese medicines have tried to improve a large number of geriatric diseases for many years. The present data suggest that HJG, one of *Kampo* drugs frequently used for treating many geriatric diseases, improves age-related unbalance of Ca metabolism. Considering that unbalanced Ca metabolism is though to be involved in dysfunction of many organs in elderly, the present finding would contribute to accumulate scientific evidence of pharmacological actions of this drug for geriatric diseases.

Conflicts of interest

None of the authors have any actual or potential conflicts of interest.

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